

## Section 8.1: Arc Length

We know that we can easily find the distance between two points; that is, it is easy to find the length of a line segment. We wonder how an arc length is defined and can be calculated.

Suppose  $f(x)$  is a function defined on  $[a, b]$  such that  $f'(x)$  is continuous. In this case, the graph of  $f$  is smooth. Let

$$P = \{x_0, x_1, \dots, x_n\}$$

be a partition of  $[a, b]$ . Let  $P_0, P_1, \dots, P_n$  be the corresponding points on the graph of  $f$ .

When  $\Delta x_i$  is “small”, the arc length  $|\widetilde{P_{i-1}P_i}|$  is close to the line segment length  $|\overline{P_{i-1}P_i}|$ :

$$\begin{aligned} |\widetilde{P_{i-1}P_i}| &\approx |\overline{P_{i-1}P_i}| = \sqrt{\Delta x_i^2 + \Delta y_i^2} \\ &= \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2} \Delta x_i \approx \sqrt{1 + f'(x_{i-1})^2} \Delta x_i. \end{aligned}$$

Thus we have an approximation of the arc length

$$L = \sum_{i=1}^n |\widetilde{P_{i-1}P_i}| \approx \sum_{i=1}^n \sqrt{1 + f'(x_{i-1})^2} \Delta x_i \longrightarrow \int_a^b \sqrt{1 + f'(x)^2} dx.$$

Therefore, we have the arc length formula

$$L = \int_a^b \sqrt{1 + f'(x)^2} dx.$$

Similarly, if  $g(y)$  is a function defined on  $[c, d]$  with  $g'(y)$  continuous, then the arc length is given by

$$L = \int_c^d \sqrt{1 + g'(y)^2} dy.$$

**Remark.** It may not be easy to calculate an arc length.

**Example 1.** Find the length of the curve  $y = 2(x+4)^{\frac{3}{2}}$  ( $0 \leq x \leq 2$ ).

Solution. Now  $y' = 3(x+4)^{\frac{1}{2}}$  and  $(y')^2 = 9(x+4)$ . Therefore, we have

$$\begin{aligned} L &= \int_0^2 \sqrt{1 + (y')^2} dx = \int_0^2 \sqrt{37 + 9x} dx = \frac{1}{9} \int_0^2 (37 + 9x)^{\frac{1}{2}} 9 dx \\ &= \frac{1}{9} \left[ \frac{2}{3} (37 + 9x)^{\frac{3}{2}} \right]_0^2 = \frac{2}{27} (\sqrt{55^3} - \sqrt{37^3}). \quad \blacksquare \end{aligned}$$

**Example 2.** Find the length of the curve  $y = x^2 - \frac{1}{8} \ln x$  ( $1 \leq x \leq 2$ ).

Solution. Now  $y' = 2x - \frac{1}{8x}$  and  $1 + (y')^2 = 4x^2 + \frac{1}{64x^2} + \frac{1}{2} = \left(2x + \frac{1}{8x}\right)^2$ .

Therefore, we have

$$L = \int_1^2 \sqrt{1 + (y')^2} dx = \int_1^2 \left(2x + \frac{1}{8x}\right) dx = \left[x^2 + \frac{1}{8} \ln x\right]_1^2 = 3 + \frac{1}{8} \ln 2. \quad \blacksquare$$

**Example 3.** Find the length of the curve  $x = \frac{y^4}{8} + \frac{1}{4y^2}$  ( $1 \leq y \leq 2$ ).

Solution. Now  $x' = \frac{1}{2}y^3 - \frac{1}{2}y^{-3}$  and  $1 + (x')^2 = \frac{1}{4}(y^6 + y^{-6} + 2) = \frac{1}{4}(y^3 + y^{-3})^2$ .

Therefore, we have

$$L = \int_1^2 \sqrt{1 + (x')^2} dy = \int_1^2 \frac{1}{2} (y^3 + y^{-3}) dy = \frac{1}{2} \left[ \frac{1}{4} y^4 - \frac{1}{2} y^{-2} \right]_1^2 = \frac{33}{16}. \quad \blacksquare$$

## Section 8.2: Area of a Surface of Revolution

We start with a simple case, where  $f$  is a linear function so that the graph of  $f$  is a line.

First, we consider the area  $A$  of a circular cone with base radius  $r$  and slant height  $\ell$ .

From  $\frac{A}{\pi \ell^2} = \frac{2\pi r}{2\pi \ell}$ , we get  $A = \pi r \ell$ .

Next, we consider the area  $A$  of a band with slant height  $\ell$  and upper and lower radii  $r_1$  and  $r_2$ . Then  $A$  can be found by subtracting the areas of the two circular cones:

$$A = A_2 - A_1 = \pi r_2(\ell_1 + \ell) - \pi r_1 \ell_1 = \pi((r_2 - r_1)\ell_1 + r_2 \ell).$$

We want to express  $\ell_1$  in terms of  $r_1$ ,  $r_2$  and  $\ell$ . From the similar triangles, we have

$$\frac{\ell_1}{r_1} = \frac{\ell_1 + \ell}{r_2} \implies r_2 \ell_1 = r_1 \ell_1 + r_1 \ell \implies (r_2 - r_1) \ell_1 = r_1 \ell.$$

Therefore, we obtain

$$A = \pi(r_1 + r_2)\ell = 2\pi r \ell, \text{ where } r = \frac{r_1 + r_2}{2}.$$

Now we consider the more general situation. Suppose  $f(x) \geq 0$  is a function defined on  $[a, b]$  such that  $f'(x)$  is continuous. Let

$$P = \{x_0, x_1, \dots, x_n\}$$

be a partition of  $[a, b]$ . Let  $P_0, P_1, \dots, P_n$  be the corresponding points on the graph of  $f$ .

When  $\Delta x_i$  is “small”, the area  $A_i$  of the band obtained by rotating the curve  $\widetilde{P_{i-1}P_i}$  about the  $x$ -axis is close to the one obtained by rotating the line segment  $\overline{P_{i-1}P_i}$  about the  $x$ -axis, and hence we have

$$\begin{aligned} A_i &\approx 2\pi f(x_{i-1}) \left| \overline{P_{i-1}P_i} \right| = 2\pi f(x_{i-1}) \sqrt{\Delta x_i^2 + \Delta y_i^2} \\ &= 2\pi f(x_{i-1}) \sqrt{1 + \left( \frac{\Delta y_i}{\Delta x_i} \right)^2} \Delta x_i \approx 2\pi f(x_{i-1}) \sqrt{1 + f'(x_{i-1})^2} \Delta x_i. \end{aligned}$$

Thus we have an approximation of the area  $A$  of the surface obtained by rotating the graph of  $f(x)$  ( $a \leq x \leq b$ ) about the  $x$ -axis

$$A = \sum_{i=1}^n A_i \approx \sum_{i=1}^n 2\pi f(x_{i-1}) \sqrt{1 + f'(x_{i-1})^2} \Delta x_i \longrightarrow \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx.$$

Therefore, we have the surface area formula

$$A = \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx.$$

If the **same** curve is described as  $x = x(y)$  for  $c \leq y \leq d$ , then the surface area is given by

$$A = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy.$$

**Example 4.** Find the area of the surface obtained by rotating about the  $x$ -axis the curve  $f(x) = \sqrt{4 - x^2}$  ( $-1 \leq x \leq 1$ ).

Solution. Now we have

$$f'(x) = \frac{-x}{\sqrt{4 - x^2}} \quad \text{and} \quad 1 + f'(x)^2 = 1 + \frac{x^2}{4 - x^2} = \frac{4}{4 - x^2}.$$

Therefore, we have

$$A = \int_{-1}^1 2\pi f(x) \sqrt{1 + f'(x)^2} dx = \int_{-1}^1 2\pi \sqrt{4 - x^2} \frac{2}{\sqrt{4 - x^2}} dx = \int_{-1}^1 4\pi dx = 8\pi.$$

Note that now the curve cannot be described as  $x = x(y)$  so that we cannot calculate the surface area by using the second formula above. ■

Similarly, if  $g(y) \geq 0$  is a function defined on  $[c, d]$  such that  $g'(y)$  is continuous, then the area of the surface obtained by rotating the graph of  $g$  about the  $y$ -axis is given by

$$A = \int_c^d 2\pi g(y) \sqrt{1 + g'(y)^2} dy.$$

If the **same** curve is described as  $y = y(x)$  for  $a \leq x \leq b$ , then the surface area is given by

$$A = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

**Example 5.** Find the area of the surface obtained by rotating about the  $y$ -axis the curve  $x = \sqrt{y}$  ( $1 \leq y \leq 4$ ).

Solution. Now  $x' = \frac{1}{2\sqrt{y}}$  and  $1 + (x')^2 = 1 + \frac{1}{4y}$ . Therefore, we have

$$\begin{aligned} A &= \int_1^4 2\pi \sqrt{y} \sqrt{1 + \frac{1}{4y}} dy = 2\pi \int_1^4 \sqrt{y + \frac{1}{4}} dy \\ &= 2\pi \left[ \frac{2}{3} \left( y + \frac{1}{4} \right)^{\frac{3}{2}} \right]_1^4 = \frac{\pi}{6} (\sqrt{17^3} - \sqrt{5^3}). \end{aligned}$$

If we describe this curve by  $y = x^2$  ( $1 \leq x \leq 2$ ), then we can also get the surface area by

$$\begin{aligned} A &= \int_1^2 2\pi x \sqrt{1 + (y')^2} dx = 2\pi \int_1^2 x \sqrt{1 + 4x^2} dx \\ &= \frac{\pi}{6} \left[ (1 + 4x^2)^{3/2} \right]_1^2 = \frac{\pi}{6} (\sqrt{17^3} - \sqrt{5^3}). \quad \blacksquare \end{aligned}$$